Problem Chosen	2026 HiMCM/MidMCM Summary Sheet	Team Control Number
B		18588

CP-SET: Measuring Useful Cooling in Open Public Spaces

Summary

An open cooling system should be judged by the useful service reaching its users. We develop the **Coupled Psychrometric-SET Service Efficiency & Effectiveness Framework (CP-SET)** to connect heat and moisture balances, local air delivery, human thermal response and operational decisions.

The freezer case establishes four essential quantities: service quantity, local quality, duration and electrical input. Its useful storage service motivates a human-service analogue: signed SET-excess relief accumulated over passenger-hours, divided by the electricity used by every operating device. This distinguishes efficiency from the adequacy of the cooling delivered.

For a New York City bus-shelter geometry, the framework combines wet-bulb-limited evaporation, dry-air-weighted local mixing and Standard Effective Temperature. At 34°C, fan A achieves efficiencies of 84.58 and 90.57 person·°C·h/kWh at 30% and 70% RH. B and C require 4.8 and 10 times A's accumulated relief to overcome their power ratios. A delivery-loss test identifies where B overtakes A. A separate hot-dry test shows why larger thermal improvement need not imply higher efficiency.

At Canberra MRT station, the same delivery and response modules are retained while zone enthalpy, humidity and surface temperature evolve with exchange, passenger loads and equipment operation. Across 24 sampled installations, A3 reaches the highest efficiency, 37.58; among six equal-coverage benchmarks, A36 leads at 35.49. Relocating the same AB30 inventory toward passenger demand raises its efficiency from 15.39 to 18.39. Coverage and local delivery therefore matter alongside the rated cooling effect.

At Ullman Terrace, Phoenix, and Satay Street, Singapore, we enumerate integer layouts and compare pure devices at equal coverage. The B–C efficiency order reverses between dry and humid climates. Adding thermal, access and annual-energy requirements yields different cost-minimizing plans: **A10C2 in Phoenix** provides 1248.87 service units using 30 kWh per four-hour session; **A10 in Singapore** provides 971.22 using 10 kWh. Their equivalent annual costs are USD 3360 and 1293 under the common planning prices. The Phoenix choice persists in 93.8% of tested price combinations; Singapore's persists in all of them.

Conservation checks, limiting-case recovery and time-step refinement verify the numerical implementation. Climate, local delivery and cost sensitivities identify the assumptions that affect the decision. The resulting memorandum and public explanation translate useful cooling per kWh into equipment placement, operating schedules and extreme-heat procedures.

Keywords: psychrometrics; local air delivery; Standard Effective Temperature; cooling service efficiency; multi-objective decisions.

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1 Introduction

1.1 Background

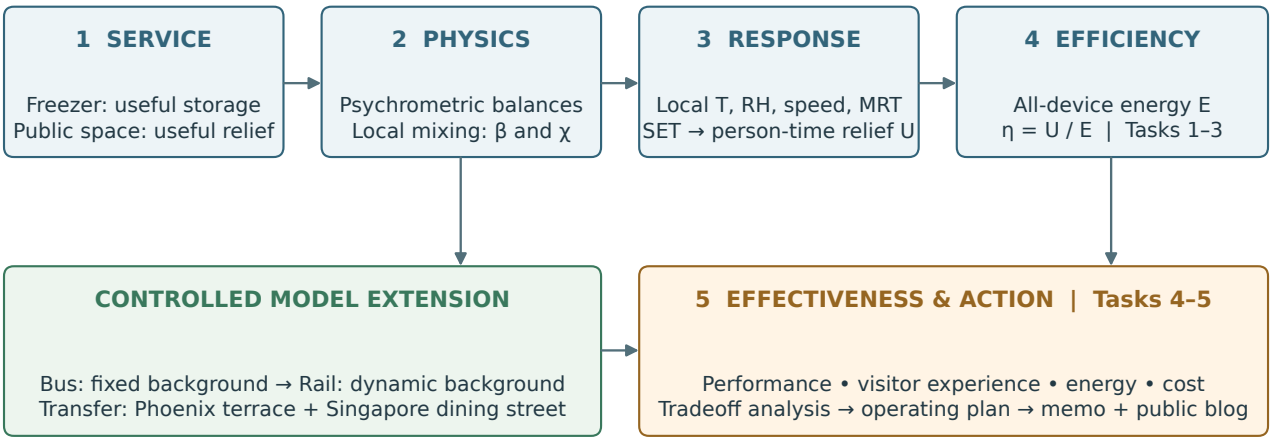
Open cooling systems must deliver relief despite continuous exchange with their surroundings. Fans alter human convective and evaporative losses; evaporative coolers also change air temperature and moisture [6, 5]. Comparing them requires useful service per unit energy.

1.2 Problem Restatement

The five tasks are to define cooling efficiency, develop bus and rail models, test two contrasting venues, extend efficiency into an effectiveness decision, and communicate operating guidance through a memorandum and public blog [4].

1.3 Our Modeling Framework

Our **CP-SET Framework (Coupled Psychrometric-SET Service Efficiency & Effectiveness Framework)** connects conserved heat and moisture to local exposure, then converts SET relief into person-time service per kWh. Requirements 1–3 compare efficiency; Requirement 4 adds cooling adequacy, visitor experience and cost. The workflow below preserves this progression.



CP-SET workflow. A shared physical and service model supports progressively broader decisions.

AI assistance in model development, coding and writing is documented in the appended report [25, 26].

2 Assumptions and Justifications

A1. Matched service boundary. Comparisons share weather, people, geometry and period; the problem's kW/hr entries are interpreted as kW. *Justification:* A common workload and complete device-energy accounting isolate strategy efficiency.

A2. Zoned air and rapid local mixing. Zones have uniform background states; local supply mixes at negligible storage, with disjoint footprints and separately prescribed speed. *Justification:* This resolves occupied-location differences when local residence time is short relative to the 120-s station disturbance.

A3. Conserved heat and water. Evaporation is near-adiabatic; background dry-air mass is fixed and exchange is balanced; shaded radiation uses mean radiant temperature. Condensation is excluded in public-space cases. *Justification:* These approximations retain the dominant transfers while counting every electrical and vapor-enthalpy source once.

A4. Reference human response. SET uses standing transport users and seated diners at 1.2 met and 0.5 clo, with a 26°C excess reference. *Justification:* One signed-relief measure compares airflow and evaporation. Dry-heat tests use $T_a = T_r = 39/40$°C and RH 15/20%: 40°C is the implementation boundary; the interior 39°C test retains the ranking [8].

A5. Explicit design inputs. Published sources establish site geometry and climate context; tabulated estimates specify operating and economic conditions. *Justification:* Reproducible inputs support sensitivity analysis. Costs use one currency and horizon; equipment clearance is the visitor-access proxy. SET relief and operational heat-safety procedures serve separate purposes.

Table 1: Transport-case inputs. Geometry sources are cited; remaining entries are model design assumptions.

Input	Bus shelter	Canberra waiting area
Geometry	4.2672×1.5240×2.7178 m [10]; three equal gross strips	L = 220 m [17]; effective waiting width 6 m, usable fraction .8, air height 4 m; three zones
Climate / radiation	T_a = 34°C; RH 30/70%; T_r = 36°C	T_o = 33°C, RH 70%; external $T_{r,o}$ = 33°C [18]
People / time	$(4, 4, 4)$ people, 1 h; redistribution $(8, 3, 1)$ or $(1, 3, 8)$	$(26.4, 44, 17.6) \rightarrow (88, 198, 66)$ linearly per 15-min cycle; 2-h warm-up + 2-h evaluation
Exchange	Fixed outdoor background	ACH $(8, 4, 10)$ plus $(8, 4, 12)$ in last 2 min; adjacent flow .88 kg dry air/s
Surface / people	Prescribed radiant conditions	Per zone C_m = 17.6 MJ/K; H_{ma}/H_{mo} = 880/1760 W/K; absorbed load 3520 W; F = (.55, .85, .55); per person q_c/q_r = 45/30 W, $g = 45/(2.45 \times 10^6)$ kg/s
Device delivery	Profiles in Table 2; $\chi = 0$; χ sensitivity uses A β = (.8, .45, .1) and rail flows	β = .5, χ = 1; A/B/C flow .6/.6/1.2 kg dry air/s; footprints 20/30/40 m^2
Shared response	Pressure 101325 Pa; v_0 = .2 m/s; body area 1.8258 m^2, external work zero; $\varepsilon_{B/C}$ = .7; outlet drops capped at 5/8°C. Θ_c sensitivity 24/26/28°C.	

Table 2: Local speed and mixing profiles in zone order; each entry describes an independent local state.

Device	Bus speed (m/s)	Bus $\beta_{s,z}$	Rail covered speed (m/s)
A	1/.55/.25	Immaterial at $\chi = 0$	.60/1/.50
B	.80/.50/.30	.80/.45/.10	.50/.80/.45
C	1/.80/.50	.90/.70/.40	.75/1/.60

Table 3: Dining-venue design assumptions. Official sources establish venue identity and climate class; numerical inputs are planning estimates. Complete inputs are in `venue_model/inputs.json`.

Scenario input	Ullman Terrace, Phoenix	Satay Street, Singapore
Venue/sector	Outdoor patio [19]; 300 m^2 in three equal zones	Open-air dining street [20]; 240 m^2 in three equal zones
Climate/guests	T_a/T_r = 36/38°C, RH 25%; n = (60, 40, 20) [21]	T_a/T_r = 32/34°C, RH 75%; n = (60, 36, 24) [18]
Delivery/response	Fixed background, $\chi = 0$; seated guests with shared met/clo/v_0; equal person fractions in three local strips using Table 2's bus profiles. Zone density uniform; footprints disjoint.	
Unit A/B/C inputs	a_k = 20/30/40 m^2; supply 1.2/2/3 kg dry air/s; clearance b_k = .5/1/1.5 m^2; capital USD 300/1000/2000; annual maintenance USD 30/100/200.	
Time and prices	H = 4 h/session, H_y = 600 h/year; life L = 5 y, r = 5%, zero salvage; USD .20/kWh and 3/m^3; water multiplier λ_w = 1.15, ρ_w = 1000 kg/m^3.	
Task 4 limits	$\overline{R}_{\max}$ = 4.5°C; $E_{\max}$ = 6000 kWh/y	$\overline{R}_{\max}$ = 7.5°C; $E_{\max}$ = 2400 kWh/y
Comparison/policy	Task 3 has no coverage constraint; its common-coverage benchmark uses q = .60. Task 4 minimizes annual cost with $q \geq .80$, $X \leq .035$ and the limits above.	
Sensitivity	Capital ±30%; water/electricity ±50%; RH ±10 pp; evaporative $\beta \pm .15$ clipped to $[0, 1]$; induced speed ±20%.	

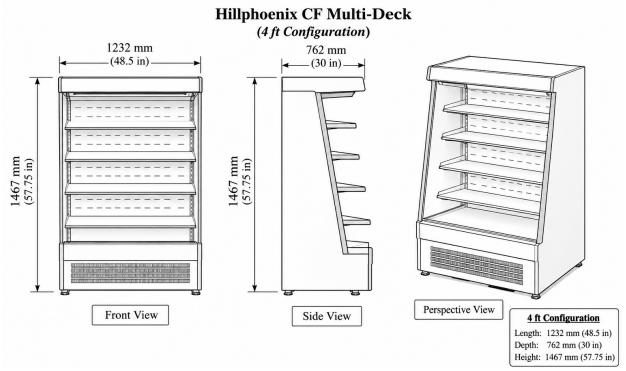
3 Nomenclature

Symbol	Meaning	Unit	Symbol	Meaning	Unit
h	Moist-air enthalpy	J/kg_{da}	w	Humidity ratio	kg/kg_{da}
$\beta_{s,z}$	Supply dry-air mixing fraction	1	χ	Pre-delivery electrical heat fraction	1
T, T_r	Air / mean radiant temperature	°C	ϕ	Relative humidity	1
Θ, Θ_c	SET / service reference	°C	D	Excess SET above Θ_c	°C
U_s	Accumulated useful relief	person·°C·h	R_s	Accumulated residual excess	person·°C·h
E_s	Electrical energy	kWh	η_s	Cooling service efficiency	person·°C·h/kWh
c_{zk}	Direct-coverage fraction	1	x_{zk}	Device count by zone/type	unit
n_z	Mean simultaneous occupants	person	$\dot{m}_k$	Dry-air supply flow	kg_{da}/s
M_z	Zone dry-air mass	kg_{da}	p_k, P_k	Device electric power	kW, W
$\overline{R}_s$	Mean residual SET excess	°C	$X_s, C_{s,y}$	Access footprint / annual cost	1, USD/year

Subscripts: s = candidate scheme, z = zone, k = device type (time interval in discrete bus sums), 0 = matched no-cooling baseline, b = background, ℓ = local exposure. Food service uses kg·h/kWh; human service uses the units above.

4 Understanding Cooling Efficiency

An open freezer succeeds by preserving food, not simply by removing heat. Low electricity use can accompany inadequate storage, while large cooling loads may reflect infiltration through the opening. We therefore evaluate *useful service per unit energy*.



The open front admits warm, moist air; cabinet surfaces conduct and radiate heat. Lights, fans, restocking and defrost add operating loads. Insulation and equipment condition modify these mechanisms; equipment age is not a separate score [1]. Product temperatures must be checked across shelves, since a mean can conceal a warm location [2].

Figure 1: Open-case heat pathways and product locations. The supplied medium-temperature display schematic illustrates geometry; freezer limits remain product-specific [11].

The numerator needs **quantity, local quality and duration**: compliant product mass integrated over storage time. The denominator includes refrigeration and auxiliaries without overlapping meter readings. Matched loading, ambient conditions and complete operating cycles prevent short operation or stored cold from inflating efficiency:

$$\boxed{\text{Cooling efficiency} = \frac{\text{useful service accumulated over objects and time}}{\text{energy input over the same period}}.} \tag{1}$$

$$\eta_f = \frac{\int \sum_i m_i(t)\mathbf{1}\{T_{p,i}(t) \in \mathcal{T}_i\}\, dt}{\int p_f(t)\, dt}, \tag{2}$$

Here m_i is product mass (kg), $\mathcal{T}_i$ its admissible temperature range, t is hours and p_f is kW; thus η_f is compliant kg·h/kWh. For people, the response changes to thermal-discomfort relief weighted by occupied time. These recipient-specific numerators cannot be directly compared numerically.

4.1 A shared physical structure

For dry-air mass M_z (kg), balanced dry-air streams $\dot{m}_{jz}$ (kg/s), enthalpy h (J/kg dry air) and humidity ratio w (kg vapor/kg dry air),

$$M_z \dot{h}_z = \sum_j \dot{m}_{jz}(h_j - h_z) + S_{h,z}, \tag{3}$$

$$M_z \dot{w}_z = \sum_j \dot{m}_{jz}(w_j - w_z) + S_{w,z}. \tag{4}$$

S_h (W) and S_w (kg/s) contain the appropriate device, occupant, product and surface sources. Freezer refrigeration can remove heat and condensate; public cooling changes airflow and moisture. The bus fixes the background and solves local delivery; the station retains that calculation while resolving background and surface storage [12]. Figure 2 shows this progression. Cruz-Salas et al. motivate separating treatment, transport and occupied-space conditions [13].

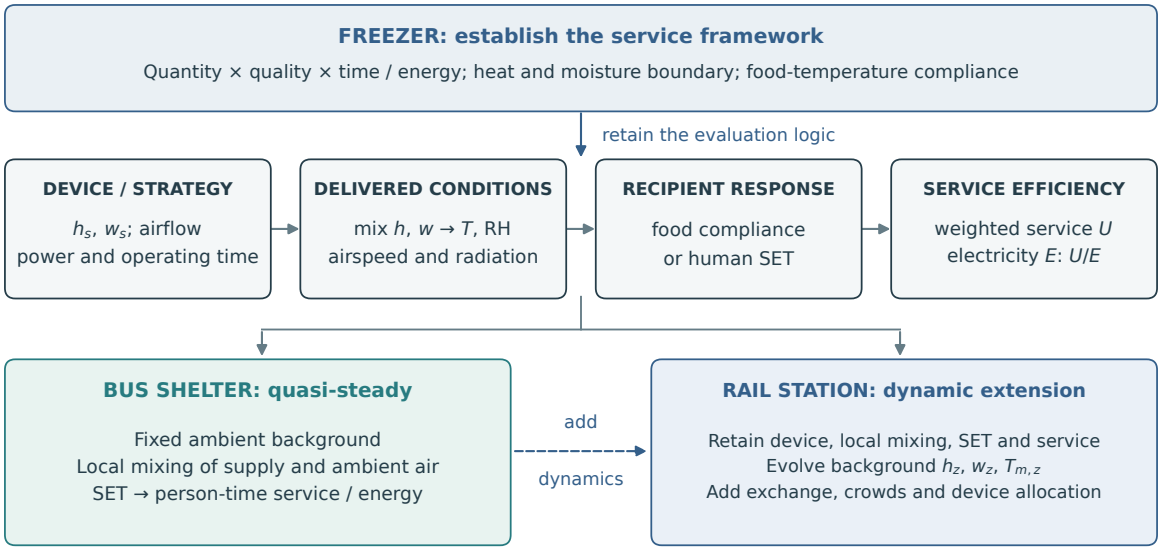


Figure 2: Shared conservation and service accounting, with recipient-specific response and increasingly resolved background dynamics.

5 Bus Stop: Local Delivery and Cooling Efficiency

5.1 Geometry and comparison inputs

NYC DOT's regular shelter measures 14 ft by 5 ft by 8 ft 11 in, or $4.2672 \times 1.5240 \times 2.7178$ m [10]. Its gross footprint is $A = 6.5032\,\mathrm{m}^2$. Three equal strips have $A_z = 2.1677\,\mathrm{m}^2$ and centers $(x_z, W/2)$, $x_z = (z - \frac{1}{2})L/3$; delivery originates at $x = 0$.

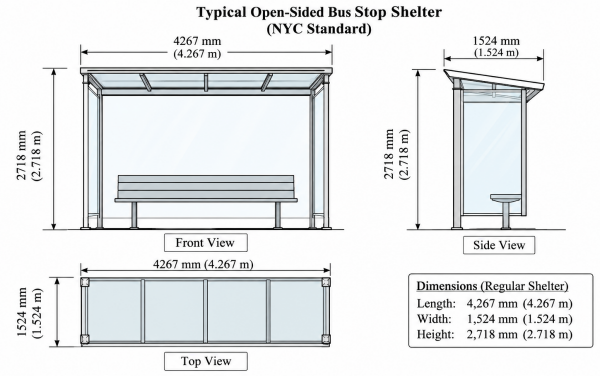


Figure 3: Adopted NYC shelter dimensions [10].

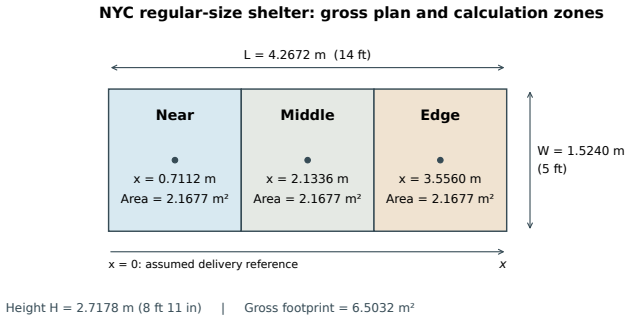


Figure 4: Equal near/middle/edge accounting strips.

The problem's 3 °C fan effect describes perceived cooling; B/C's 5/8 °C describe air cooling [4]. We interpret the listed kW/hr numbers as powers $p_{A/B/C} = 0.25/1.2/2.5$ kW. Tables 1–2 specify the comparison: 34 °C, RH 30/70%, $T_r = 36$ °C and 12 mean occupants. Geometry enters once through $n_z = \rho_z A_z$ when density replaces headcount.

5.2 From device outlet to occupied air

A changes prescribed local speed while retaining background temperature and moisture in the zero-local-heat base case [6]. For B/C, wet-bulb temperature bounds evaporation [5]:

$$T_{o,s} = T_b - \min\{\Delta T_s^{\max},\ \varepsilon_s(T_b - T_{wb,b})\}, \qquad s \in \{B, C\}, \tag{5}$$

b denotes the inlet background; ε_s is evaporative effectiveness and $\Delta T_{B/C}^{\max} = 5/8\,^\circ\mathrm{C}$ are the adopted outlet caps. With ϕ as RH fraction and pressures in Pa, SI psychrometrics give [9]

$$w = 0.621945\frac{p_v}{p - p_v}, \qquad p_v = \phi p_{sat}(T), \tag{6}$$

$$h(T, w) = 1006T + w(2501000 + 1860T), \tag{7}$$

Temperatures are in °C. The near-isenthalpic evaporative stage, $h_{o,s} \simeq h_b$, yields

$$w_{o,s} = \frac{h_b - 1006T_{o,s}}{2501000 + 1860T_{o,s}}. \tag{8}$$

This stage precedes electrical heating and neglects make-up-water enthalpy [3]. A device supply mixes with entrained background in a negligible-storage local region. For $x \in \{h, w\}$,

$$0 = \dot{m}_{s,z}(x_s^{\mathrm{sup}} - x_{\ell,s,z}) + \dot{m}_{e,z}(x_b - x_{\ell,s,z}), \qquad \beta_{s,z} = \frac{\dot{m}_{s,z}}{\dot{m}_{s,z} + \dot{m}_{e,z}}. \tag{9}$$

β is a dry-air flow fraction, not a fraction of passengers or total flow assigned to a strip. A fraction χ of unit electric power $P_s = 1000p_s$ W heats unit supply flow $\dot{m}_s$ before reaching people:

$$h_s^{\mathrm{sup}} = h_{o,s} + \chi P_s/\dot{m}_s, \qquad w_s^{\mathrm{sup}} = w_{o,s}, \qquad 0 \le \chi \le 1. \tag{10}$$

For A, the unheated outlet equals the inlet. Mixing conserves h, w, after which temperature and RH are recovered:

$$w_{\ell,s,z} = (1 - \beta_{s,z})w_b + \beta_{s,z}w_s^{\mathrm{sup}}, \tag{11}$$

$$h_{\ell,s,z} = (1 - \beta_{s,z})h_b + \beta_{s,z}h_s^{\mathrm{sup}}, \tag{12}$$

$$T_{\ell,s,z} = \frac{h_{\ell,s,z} - 2501000w_{\ell,s,z}}{1006 + 1860w_{\ell,s,z}}. \tag{13}$$

RH is not averaged directly. The bus uses $b = a$ and $\chi = 0$. Its B/C fractions imply entrainment-to-supply ratios $(1 - \beta)/\beta = 0.25/1.22/9.00$ and $0.11/0.43/1.50$. Parallel strip branches share one device's total supply.

5.3 A common response and efficiency measure

Standard effective temperature (SET) combines air temperature, radiation, airspeed and humidity for the reference passenger [7, 8]:

$$\Theta_{s,z} = F_{\mathrm{SET}}(T_{\ell,s,z}, T_{r,z}, v_{s,z}, \phi_{\ell,s,z}; \mathrm{met}, \mathrm{clo}). \tag{14}$$

For the assumed hot-discomfort reference $\Theta_c = 26\,^\circ\mathrm{C}$, define

$$D(\Theta) = [\Theta - \Theta_c]_+, \qquad [x]_+ = \max(x, 0). \tag{15}$$

Against the same shelter without active cooling, the signed improvement is

$$b_{s,z,k} = D(\Theta_{0,z,k}) - D(\Theta_{s,z,k}). \tag{16}$$

Adverse effects remain negative. With $n_{z,k}$ people and time steps Δt_k in hours,

$$U_s = \sum_k \sum_z n_{z,k} b_{s,z,k} \Delta t_k, \qquad E_s = \sum_k p_{s,k} \Delta t_k, \tag{17}$$

$$\eta_s = U_s / E_s, \qquad E_s > 0. \tag{18}$$

Here E is kWh and η is person·°C·h/kWh, not COP. Empty zones earn no service; the zero-energy baseline has no ratio. The residual provides an accounting check:

$$R_s = \sum_k \sum_z n_{z,k} D(\Theta_{s,z,k}) \Delta t_k, \qquad \overline{R}_s = \frac{R_s}{\sum_k \sum_z n_{z,k} \Delta t_k}. \tag{19}$$

Thus $U_s + R_s = R_0$. We rank efficiency by η; overall adequacy enters Section 8. Under common operating time, the power ratios give delivery-independent break-even conditions:

$$\begin{aligned} \eta_B > \eta_A &\iff U_B > 4.8 U_A, \\ \eta_C > \eta_A &\iff U_C > 10 U_A, \\ \eta_C > \eta_B &\iff U_C > \frac{25}{12} U_B. \end{aligned} \tag{20}$$

5.4 Results: humidity and spatial delivery

For C, dry-case SET rises from 26.3 °C near the outlet to 29.4 °C at the edge; at RH 70%, it rises from 33.1 to 35.2 °C. These gradients explain why a single air-temperature drop misses delivered service.

Dry-case efficiencies are 84.6/24.5/21.1 for A/B/C; humid-case values are 90.6/21.1/15.2 (Figure 6). B/C outlet drops fall from 5/8 °C to about 3.4/3.4 °C, as their common wet-bulb limit replaces the caps.

Moving the same 12 people toward the edge lowers dry-case A efficiency from 118.3 to 48.3 and reverses B/C ordering: 24.5 > 21.1 under uniform occupancy becomes 15.5 < 16.6 under edge-heavy occupancy. A leads all six nominal climate–occupancy cases.

5.5 Delivery failure and robustness

Aiming or obstruction can weaken delivered airflow without reducing power. Holding $p_A = 0.25$ kW, test

$$v_{A,z}(q) = v_0 + q(v_{A,z}^{\mathrm{nom}} - v_0), \qquad 0 \le q \le 1, \tag{21}$$

q is a delivery factor, not a motor-speed setting. On the sampled grid, A first wins at $q = 0.18$ for RH 30% and 0.16 for RH 70%; the threshold lies within the preceding 0.02 interval. B wins below it.

One-factor endpoint tests give maximum B/A and C/A efficiency ratios 0.432 and 0.372. Including full local electric heating ($\chi = 1$) changes dry-case efficiencies to 81.0/20.8/18.0 and humid values to 86.8/17.4/12.1; A remains first under those inputs. Psychrometric round trips, mass/enthalpy closure, $U + R = R_0$, zero-occupancy and signed-adverse checks pass. The delivery-loss reversal identifies a stronger sensitivity than these individual endpoint perturbations.

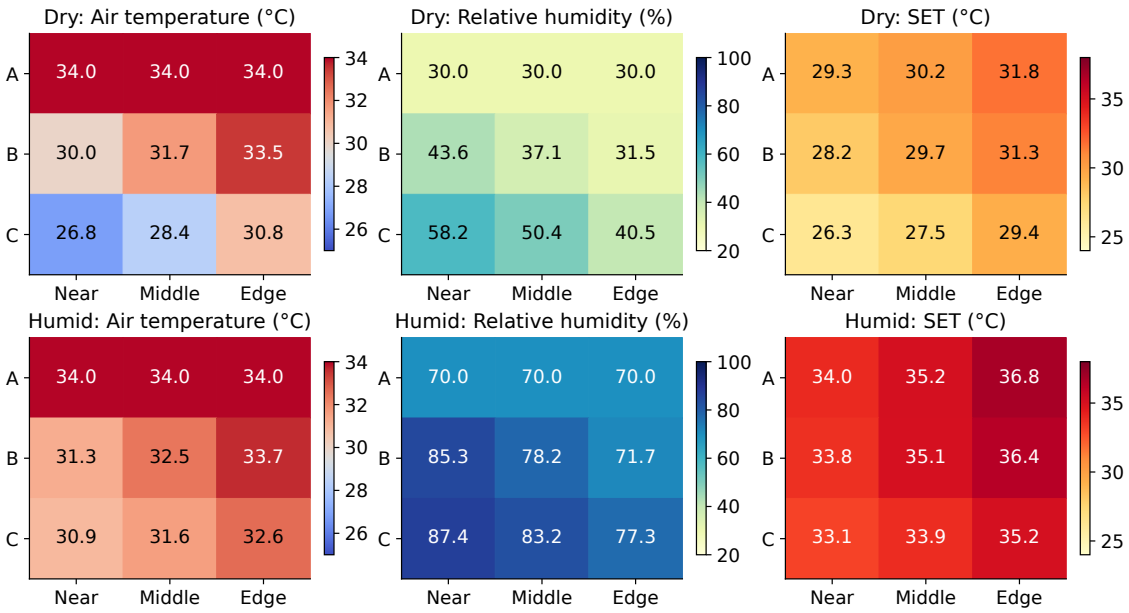


Figure 5: Local conditions under the bus inputs: A changes SET through airflow; B/C also change temperature and moisture.

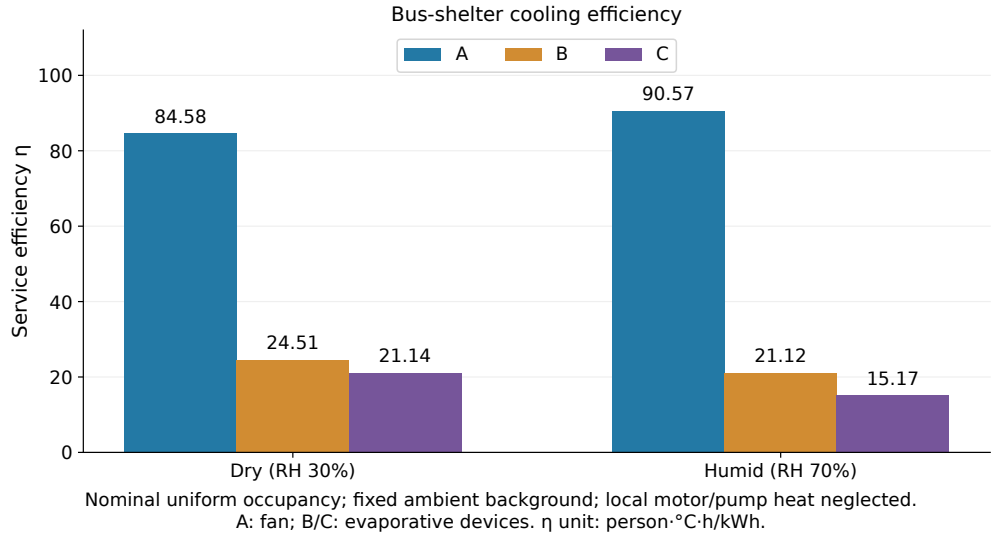


Figure 6: Uniform-occupancy efficiency: A leads both nominal humidity cases.

5.6 Extreme dry heat: testing the efficiency ranking

We test $T_a = T_r = 40\,^{\circ}\mathrm{C}$ at RH 15% and 20%, retaining the device powers, delivery profiles, 5/8 °C outlet caps, and uniform occupancy. Figure 9 shows $\eta_A > \eta_B > \eta_C$ in both cases. At RH 15%, the mean SET reductions are 1.43, 1.95, and 3.42 °C, whereas the efficiencies are 68.43, 19.48, and 16.41 person·°C·h/kWh. C produces greater relief per passenger, but consumes ten times A's electricity. SET relief explains the service numerator; η remains the comparison criterion.

The heat balance explains why dry-bulb cooling alone cannot establish the ranking. When air is warmer than the exposed skin or clothing surface, faster flow increases convective heat gain; it also promotes evaporative heat loss. SET combines these competing exchanges rather than imposing a temperature-triggered reversal of fan performance [7].

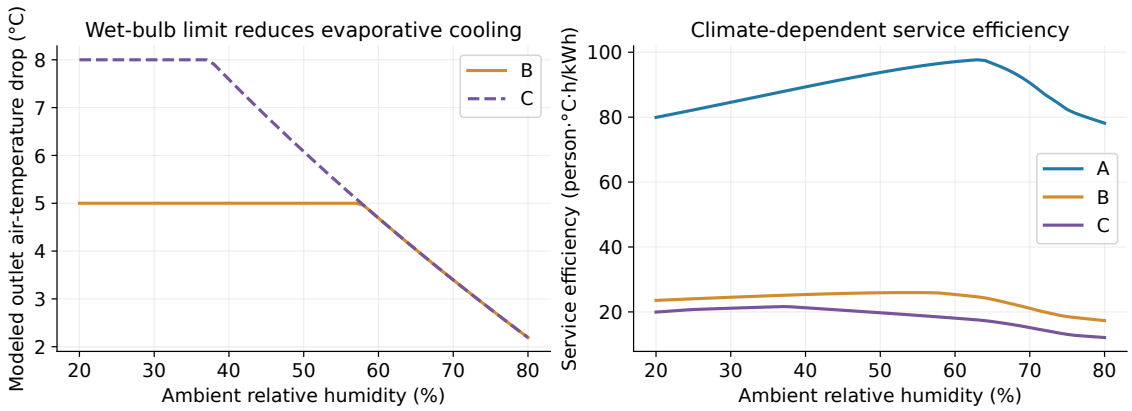


Figure 7: Humidity changes evaporative potential and SET-based efficiency at fixed delivery and demand.

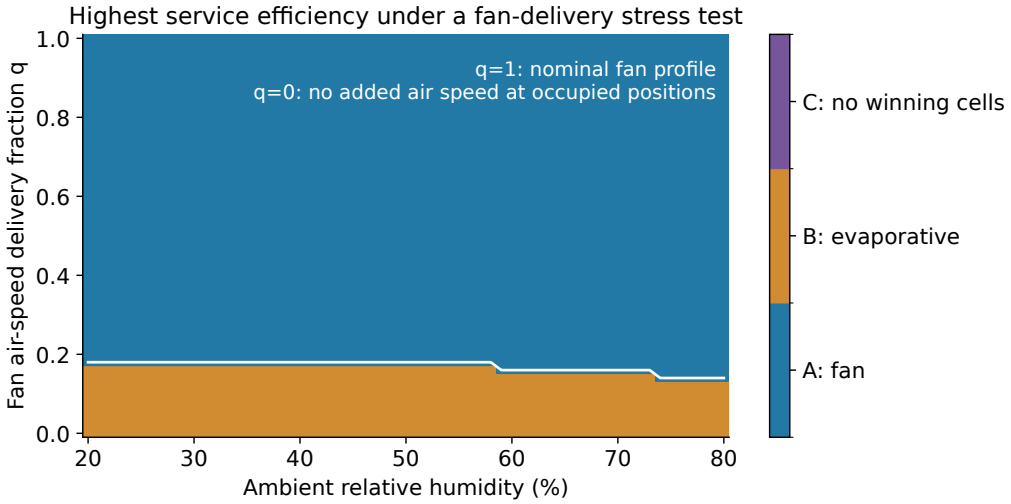


Figure 8: Efficiency winner for RH 20–80% and delivered-speed factor q; C wins no cell of this grid.

6 Canberra Station: Dynamic Background and Local Service

6.1 From a fixed reservoir to coupled zones

Canberra MRT is an elevated, naturally ventilated station in Singapore [15, 16]. Using its approximate $L = 220$ m length with assumed aggregate waiting width $b = 6$ m and usable fraction $\alpha = 0.8$ gives

$$A_{\text{occ}} = \alpha L b = 1056\,\text{m}^2. \tag{22}$$

Three access/core/edge zones each have 352 m^2 and, at 4 m effective height, 1408 m^3. Table 1 defines the 33 °C, RH 70% comparison. Its 4–10 h^{-1} background exchange implies 6–15 min replacement times, comparable to the 15-min passenger cycle: storage and humidity feedback must now be resolved.

Each zone retains h_z, w_z and effective surface temperature $T_{m,z}$, with constant reference dry-air mass $M_z = \rho_{da,o} V_z$. The device, local-mixing and SET operators remain those of the bus. For $x = h$ or w, balanced exchange is

$$\mathcal{L}_z(x) = \dot{m}_{0z}(x_o - x_z) + \sum_{j \neq z} \dot{m}_{zj}(x_j - x_z), \qquad \dot{m}_{0z} = M_z \lambda_z / 3600. \tag{23}$$

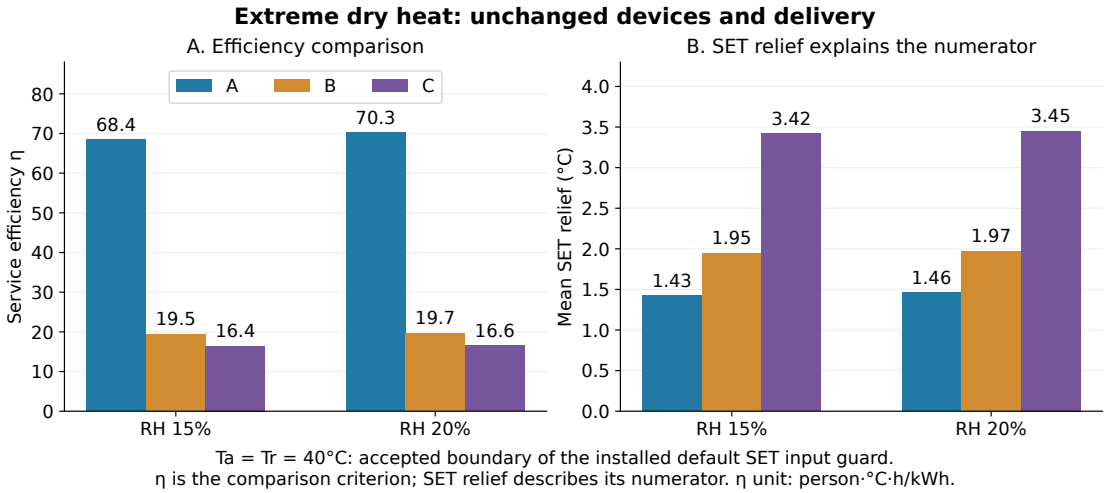


Figure 9: Extreme dry-heat boundary test with the original device powers and delivery profiles. The left panel compares cooling efficiency; the right panel explains its SET-relief numerator. Inputs follow Assumption A4.

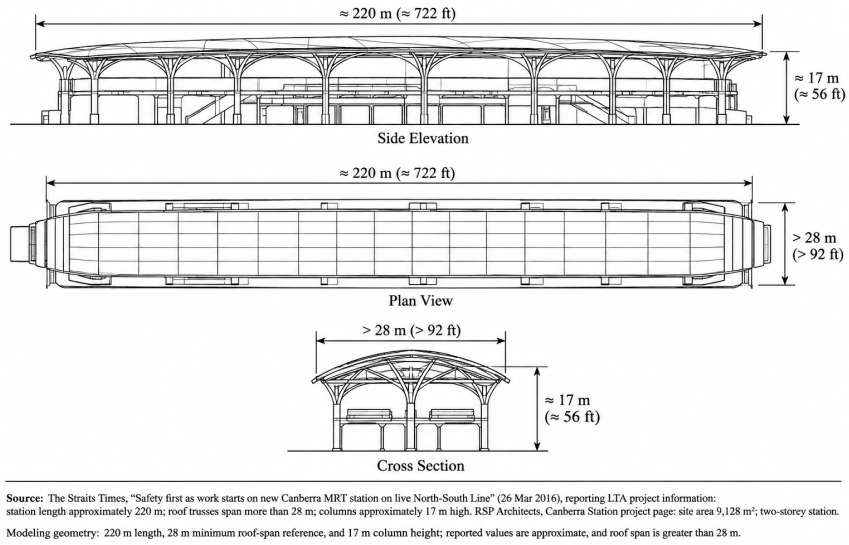


Figure 10: Canberra station context. Structural dimensions guide scale; the three waiting zones are modeled separately [17, 15].

λ_z is ACH in h^{-1}; interzone rates satisfy $\dot{m}_{zj} = \dot{m}_{jz}$. Outdoor exchange represents wind/opening effects through the prescribed cycle [14]. With time in seconds, the three conservation equations per zone are [12]

$$M_z \dot{h}_z = \mathcal{L}_z(h) + H_{ma,z}(T_{m,z} - T_z) + n_z q_c + n_z g h_v(T_z) + P_z + \sum_k x_{zk} \dot{m}_k (h_{e,zk} - h_z), \tag{24}$$

$$M_z \dot{w}_z = \mathcal{L}_z(w) + n_z g + \sum_k x_{zk} \dot{m}_k (w_{e,zk} - w_z), \tag{25}$$

$$C_{m,z} \dot{T}_{m,z} = H_{mo,z}(T_o - T_{m,z}) + Q_{\mathrm{sol},z} + n_z q_r - H_{ma,z}(T_{m,z} - T_z). \tag{26}$$

Here n_z is occupancy; per-person q_c, q_r are convective/radiant W and g is vapor kg/s. Vapor enthalpy is $h_v(T) = 2501000 + 1860T$ J/kg, with no second latent term. H_{ma}, H_{mo} are W/K, C_m is J/K, Q_{sol} is W,

and $P_z = \sum_k x_{zk} P_k$ with $P_k = 1000 p_k$ W. Subscript e denotes the unheated device outlet (the bus's o, s), so A has no evaporative source; B/C add moisture while approximately conserving outlet enthalpy. Surface exchange cancels between the air and wall budgets.

Recover $T_z = (h_z - 2501000 w_z)/(1006 + 1860 w_z)$; effective view fraction F_z determines mean radiant temperature [7]:

$$T_{r,z} = \left[F_z (T_{m,z} + 273.15)^4 + (1 - F_z)(T_{r,o} + 273.15)^4\right]^{1/4} - 273.15. \tag{27}$$

The separate outdoor radiant input is $T_{r,o}$. Increasing w_z reduces the next device pass's wet-bulb depression, providing the feedback absent from the fixed-background bus.

6.2 Local delivery with exact return accounting

Device flow and entrained background return to their original zone. For finite entrainment ($\beta > 0$), the local mixing equations imply

$$(\dot{m}_k + \dot{m}_e)(h_{\ell,zk} - h_z) + (1 - \chi) P_k = \dot{m}_k (h_{e,zk} - h_z) + P_k, \tag{28}$$

$$(\dot{m}_k + \dot{m}_e)(w_{\ell,zk} - w_z) = \dot{m}_k (w_{e,zk} - w_z). \tag{29}$$

Thus the local exposure calculation preserves the existing net heat and moisture sources exactly. Covered people use local T, ϕ, v; uncovered people use background T, ϕ, v_0, and both share T_r. Here $\beta = 0.5$, $\chi = 1$ and A/B/C unit flows are $0.6/0.6/1.2$ kg dry air/s under Table 1; $\chi = 1$ places all electrical heat in supply before passengers.

Two reductions identify the inheritance precisely. At $\beta = 0$, the scalar exposure becomes the original shared-background calculation while prescribed speeds remain. Fixing background to outdoor h_a, w_a, using bus radiation, speeds, occupancy and β, and setting $\chi = 0$ recovers the bus calculation exactly. Setting time derivatives to zero alone would not impose that reservoir boundary.

6.3 Counts, coverage and efficiency

Let s index an installation and x_{zk} its nonnegative integer count of type k in zone z. For unit footprints $a_{A/B/C} = 20/30/40\,\mathrm{m}^2$, uniform zone density gives

$$c_{zk} = \frac{x_{zk} a_k}{A_z}, \qquad c_{z0} = 1 - \sum_k c_{zk} \geq 0, \qquad \sum_k x_{zk} a_k \leq A_z. \tag{30}$$

These passenger fractions c differ from the air-mixing fraction β. Average discomfort across subgroups, not velocities or device efficiencies:

$$\overline{D}_{s,z}(t) = c_{z0} D[\Theta(T_z, T_{r,z}, v_0, \phi_z)] + \sum_k c_{zk} D[\Theta(T_{\ell,zk}, T_{r,z}, v_{zk}, \phi_{\ell,zk})]. \tag{31}$$

All units affect the background; direct coverage determines the additional local exposure. Every plan starts after the same two-hour uncooled warm-up, runs two hours, and uses a matched no-device baseline. The common passenger history totals 440 person-hours. With p_k in kW,

$$U_s = \frac{1}{3600} \int_0^{7200} \sum_z n_z(t) [D_{0,z}(t) - \overline{D}_{s,z}(t)]\, dt, \tag{32}$$

$$E_s = \frac{1}{3600} \int_0^{7200} \sum_{z,k} x_{zk} p_k\, dt, \qquad \eta_s = U_s / E_s. \tag{33}$$

The $1/3600$ factors convert seconds to hours, retaining person·°C·h/kWh. This section compares U/E. Mean residual $\overline{R}_s = \int \sum_z n_z \overline{D}_{s,z}\, dt / \int \sum_z n_z\, dt$ supplies the later adequacy criterion in Section 8.

6.4 Deployment results and verification

Among 24 sampled plans, A3 maximizes η at 37.58; among the six equal-68.18%-coverage benchmarks, A36 leads at 35.49. The distinct candidate sets answer scale and device-comparison questions, respectively (Figure 11).

Plan suffixes give total device counts. For A36/B24/C18/AB30/AC27/ABC24, per-zone (A, B, C) counts are $(12, 0, 0)$, $(0, 8, 0)$, $(0, 0, 6)$, $(6, 4, 0)$, $(6, 0, 3)$ and $(3, 2, 3)$, respectively; A3 places one fan in each zone.

Table 4: Nominal efficiency accounting: U in person·°C·h, E in kWh.

Plan	Person-time coverage	U	E	η
A3	5.68%	56.38	1.5	37.58
A36	68.18%	638.89	18.0	35.49
B24	68.18%	446.28	57.6	7.75
C18	68.18%	453.00	90.0	5.03
AB30	68.18%	581.61	37.8	15.39
AC27	68.18%	593.68	54.0	10.99
ABC24	68.18%	524.76	63.9	8.21
AB30-R	76.19%	695.07	37.8	18.39

Redistributing AB30's 18 A and 12 B units to zone counts $(3, 4, 0)$, $(12, 3, 0)$ and $(3, 5, 0)$ raises person-time coverage from 68.18% to 76.19%. At the same 37.8 kWh, U rises from 581.61 to 695.07 and η from 15.39 to 18.39: allocation changes useful delivery without adding equipment.

A36 retains its equal-coverage efficiency lead for all 15 combinations $\beta = 0/0.25/0.5/0.75/1$, $\chi = 0/0.5/1$. Varying A's unit flow through $0.3/0.6/1.2$ kg dry air/s gives efficiencies $34.06/35.49/36.22$. Figure 12 shows the controlled delivery sensitivity.

Global energy/moisture budgets and local return identities close numerically. Halving the time step changes the tested mixed-plan efficiencies by at most 5.4×10^{-6} relatively; $\beta = 0$ reproduces all 24 former shared-background results exactly. The fixed-reservoir test recovers bus states and service. These checks verify the extension's conservation and reduction properties.

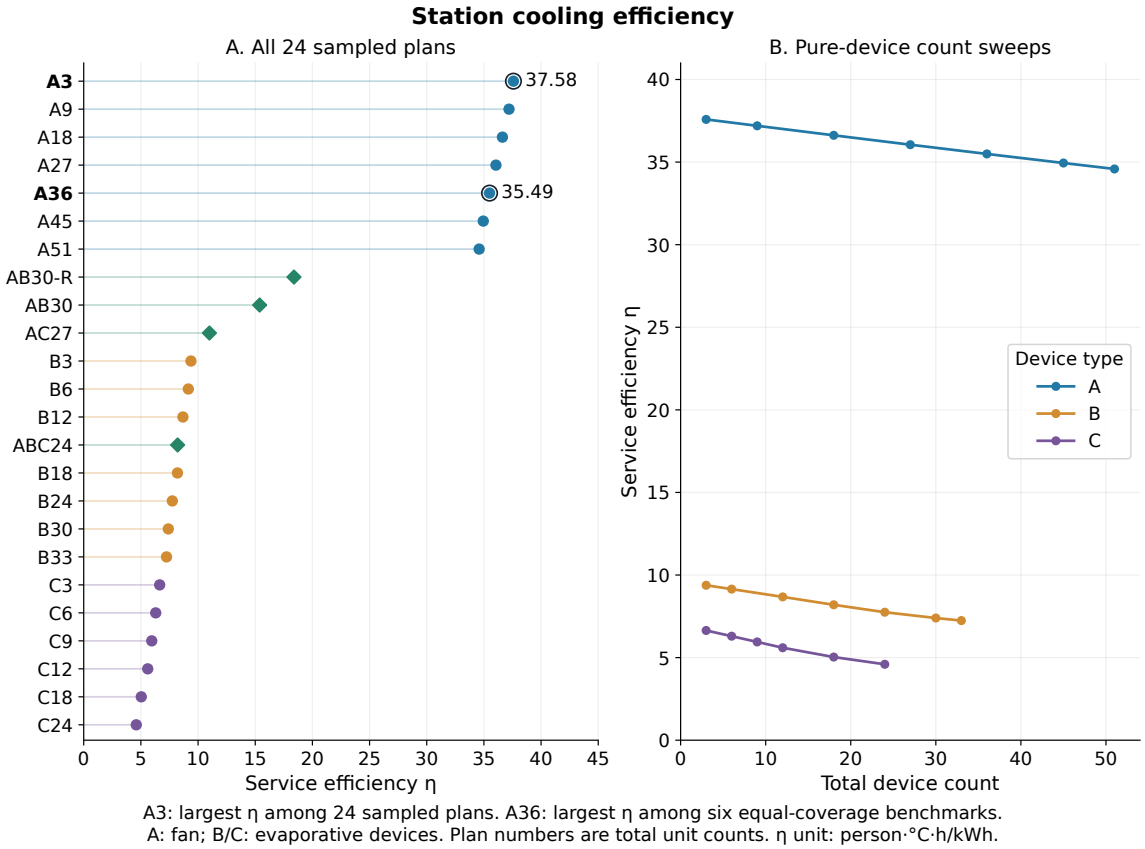


Figure 11: A3 leads all 24 sampled installations; A36 leads the six equal-coverage benchmarks. Device counts change both delivered service and background loads.

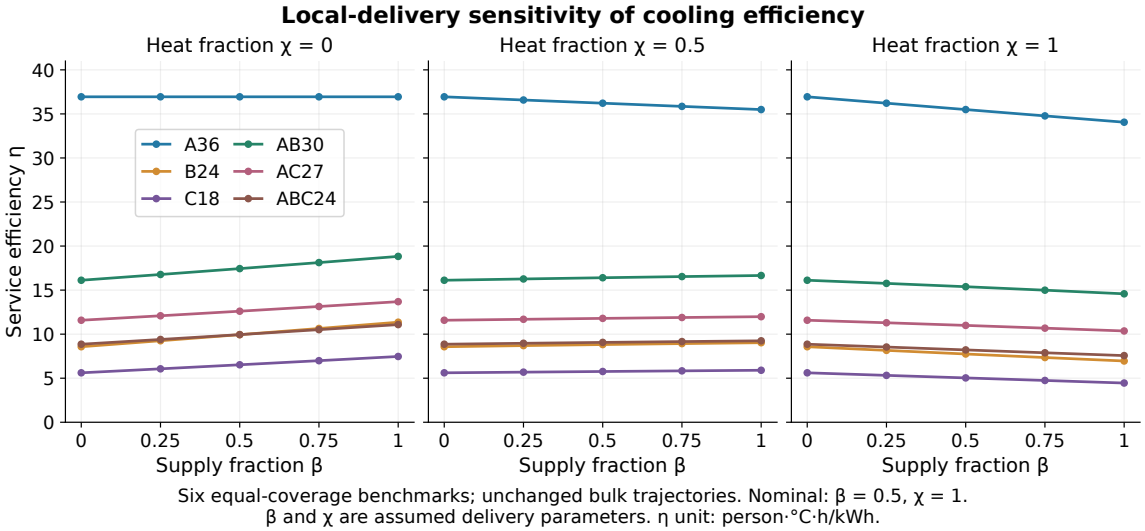


Figure 12: Efficiency sensitivity to mixing and heat allocation; background trajectories remain unchanged by the local mapping.

7 Apply the Model: Two Outdoor Dining Venues

7.1 Different climates, the same service calculation

We apply CP-SET to a dining sector of **Ullman Terrace at Desert Botanical Garden, Phoenix**, an outdoor event patio [19], and **Satay Street at Lau Pa Sat, Singapore**, an open-air dining street on Boon Tat Street [20]. Their dry-desert and humid-tropical climate contrast is represented by the warm-day scenarios in Table 3.

Both spaces are open enough for a prescribed-background planning approximation. We retain the bus model's device outlets, enthalpy–moisture mixing, local-speed profiles and SET calculation, together with the station model's integer allocation and disjoint coverage. Three delivery strips within each device footprint represent attenuation from its near to far edge. Their equal-weight mean excess is $D_k = \frac{1}{3}\sum_{j=1}^{3} D(\Theta_{kj})$. With the coverage fractions c_{zk} from Eq. (30),

$$\overline{R}_s = \frac{\sum_z n_z\,[(1-\sum_k c_{zk})D_0 + \sum_k c_{zk}D_k]}{\sum_z n_z}, \qquad U_s = H\Big(\sum_z n_z\Big)(D_0 - \overline{R}_s). \tag{34}$$

Thus geometry controls who receives relief, while $E_s = H\sum_{z,k} x_{zk}p_k$ counts every device. The supply flow used for water demand belongs to each physical device, not separately to each delivery strip. Fixed background and $\chi = 0$ retain the shelter approximation; crowd-driven accumulation or poorly flushed enclosures require the dynamic station equations instead.

7.2 Climate changes the value of evaporation

Task 3 ranks η over every positive-energy layout satisfying nonoverlap: 12,166 in Phoenix and 3,374 in Singapore. **The maximum efficiencies are** 81.10 **and** 134.89 **person·°C·h/kWh.** A single fan in the busiest zone attains each maximum, covering 10% and 12.5% of guest-time, respectively. Up to five such fans in Phoenix and four in Singapore tie while remaining inside that zone; no coverage, cost or residual threshold enters this efficiency ranking.

Table 5 separately compares pure types at the same coverage. Within each type, we choose the least-energy layout achieving that coverage. The wet-bulb temperatures are 20.90°C in Phoenix and 28.16°C in Singapore. Consequently, C achieves its 8°C outlet-drop cap in the dry scenario, but only 2.68°C in the humid scenario. Dilution and the SET response translate this outlet change into useful service.

Table 5: Equal-coverage efficiency comparison: every plan serves 60% of guest-time. U is person·°C·h and $\eta = U/E$.

Venue	Plan	U	E (kWh)	η
Phoenix	A7	486.58	7.0	69.51
	B5	666.41	24.0	27.77
	C4	1183.98	40.0	29.60
Singapore	A6	647.48	6.0	107.91
	B4	728.44	19.2	37.94
	C3	1069.98	30.0	35.67

Fans give the greatest service per kWh in both comparisons. The B–C efficiency order reverses: C exceeds B in Phoenix, whereas B exceeds C in Singapore. Humidity removes C's larger outlet-

temperature-drop advantage, leaving delivery reach and unit power to determine its efficiency. Singapore's larger fan efficiency also reflects its denser dining sector and stronger modeled SET response to air movement. Occupancy, delivery and climate therefore belong in the same calculation.

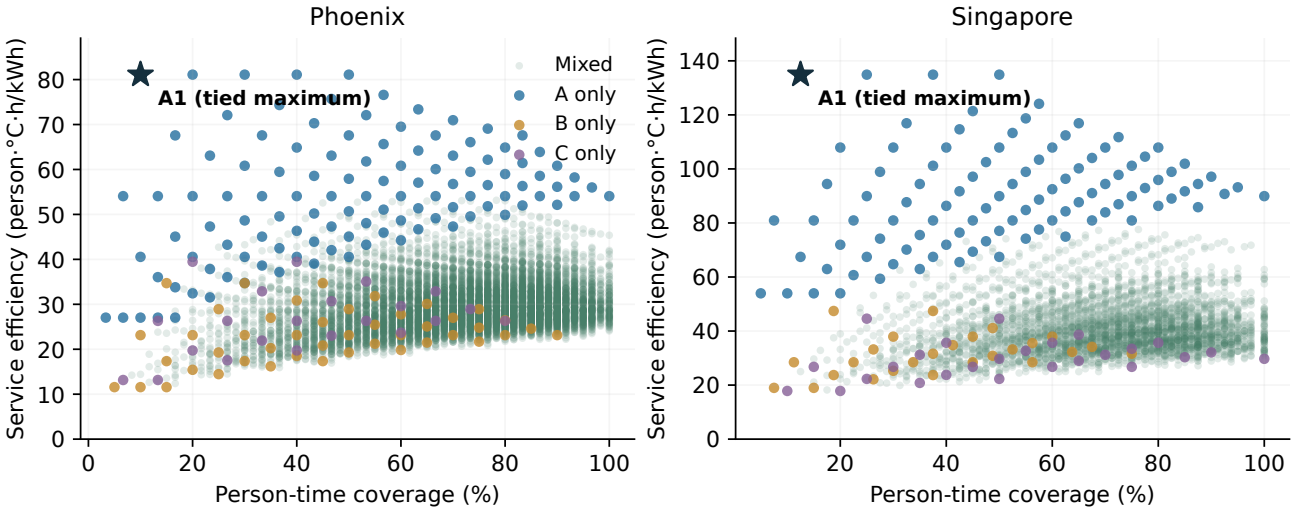


Figure 13: Efficiency over all positive-energy integer layouts, with no coverage threshold. Coincident points are merged for display. Stars identify the smallest installations attaining the tied maximum; broader service changes the relevant occupied locations.

7.3 What scales, and what does not

Efficient cooling depends chiefly on delivering air to occupied locations, preserving useful speed, limiting dilution, and exploiting a sufficient dry-bulb–wet-bulb difference. Increasing area changes the equipment count and occupied-location weights; it does not multiply η directly.

A scale check makes this distinction concrete. Doubling Singapore's area and guests at the same density, while retaining the reference fan layout $(4, 4, 2)$, leaves $U/E = 97.12$ unchanged but halves coverage from 90% to 45%. Doubling the devices as well restores coverage, doubles electricity from 10 to 20 kWh, and again preserves U/E. Efficiency is therefore an intensity measure; service adequacy enters the next task. The code verifies these scaling identities, $U + R = R_0$, saturation bounds and the shared heat/moisture conservation checks.

8 Overall Effectiveness: Four Objectives, an Explicit Choice

8.1 Performance, access, energy and cost

We retain the physical outputs and compare four quantities,

$$\min_s \left(\overline{R}_s,\ X_s,\ E_{s,y},\ C_{s,y}\right), \qquad X_s = \frac{\sum_{z,k} x_{zk} b_k}{\sum_z A_z}. \tag{35}$$

Remaining SET excess measures thermal performance. X_s is the fraction of dining area reserved for equipment and maintenance clearance, providing a calculable *visitor-access proxy*. Clear walkways and exits remain layout requirements. $E_{s,y}$ measures resource use, and $C_{s,y}$ measures annual expenditure. Thermal relief is counted once; η is not a fifth objective.

All ownership, maintenance and running costs use the annual basis in Table 3. With capital recovery factor $\mathrm{CRF} = r(1+r)^L/[(1+r)^L - 1]$,

$$\begin{aligned} C_{s,y} &= \mathrm{CRF} \sum_{z,k} x_{zk} K_k + \sum_{z,k} x_{zk} M_k + \pi_e E_{s,y} + \pi_w V_{s,y}, \\ V_{s,y} &= \frac{3600 H_y \lambda_w}{\rho_w} \sum_{z,k} x_{zk} \dot{m}_k (w_{e,k} - w_a). \end{aligned} \tag{36}$$

Here K_k and M_k are initial and annual maintenance costs, H_y is annual operating hours, and ρ_w converts evaporated mass to water volume. The common USD price basis supports comparison. EIA's Arizona commercial average, USD 0.1245/kWh in 2025 [22], falls within the electricity sensitivity range.

8.2 Conditional recommendations from the full candidate set

For Task 4, we impose the assumed coverage requirement, then remove four-objective dominated options. Phoenix/Singapore have 199/106 eligible count combinations and 80/32 **coverage-constrained Pareto options**. For identical counts, energy, cost and clearance are position-independent here; retaining the eligible layout with least residual therefore preserves the policy optimum. Under an illustrative operator policy,

$$s^* = \arg\min_s C_{s,y} \quad \text{subject to} \quad q_s \geq q_{\min}, \quad \overline{R}_s \leq \overline{R}_{\max}, \quad X_s \leq X_{\max}, \quad E_{s,y} \leq E_{\max}. \tag{37}$$

The limits in Table 3 express a budgeted service choice without arbitrary equal weighting. Exhaustive enumeration establishes the minimum within the discrete design space.

Table 6: Selected plans under the stated four-objective policy. Annual costs include ownership, maintenance, electricity and water.

Venue	Plan	q (%)	$\overline{R}$ (°C)	X (%)	E_y (kWh)	C_y (USD)
Phoenix	A10C2	96.67	4.482	2.67	4500	3360
Singapore	A10	90.00	7.453	2.08	1500	1293

In Phoenix, place two C units and one A in the busiest zone, five A units in the second, and four A in the third. This A10C2 layout delivers 1248.87 person·°C·h per session using 30 kWh and 0.319 m^3 of make-up water. A15 is cheaper but misses the residual requirement; C6 exceeds the energy budget. In Singapore, the A allocation $(4, 4, 2)$ meets the policy with 971.22 service units and 10 kWh. Its ten fans require no evaporative make-up water. The different choices emerge from the constraints and climate response.

8.3 Sensitivity leads to an operating policy

Across 243 price combinations, Phoenix retains A10C2 in 228 (93.8%) and selects A4C3 in 15; Singapore retains A10 throughout. Physical changes have stronger effects in these tests. A 0.15 loss in evaporative delivery fraction selects A6C3 in Phoenix; a 20% loss in induced speed selects A7C3 there and A12 in Singapore. Raising RH by ten percentage points leaves both venues without a feasible layout under their unchanged service, access and energy policies. Operators should then revise timing, radiant exposure or the service budget and reoptimize. The memorandum translates these triggers into operating actions.

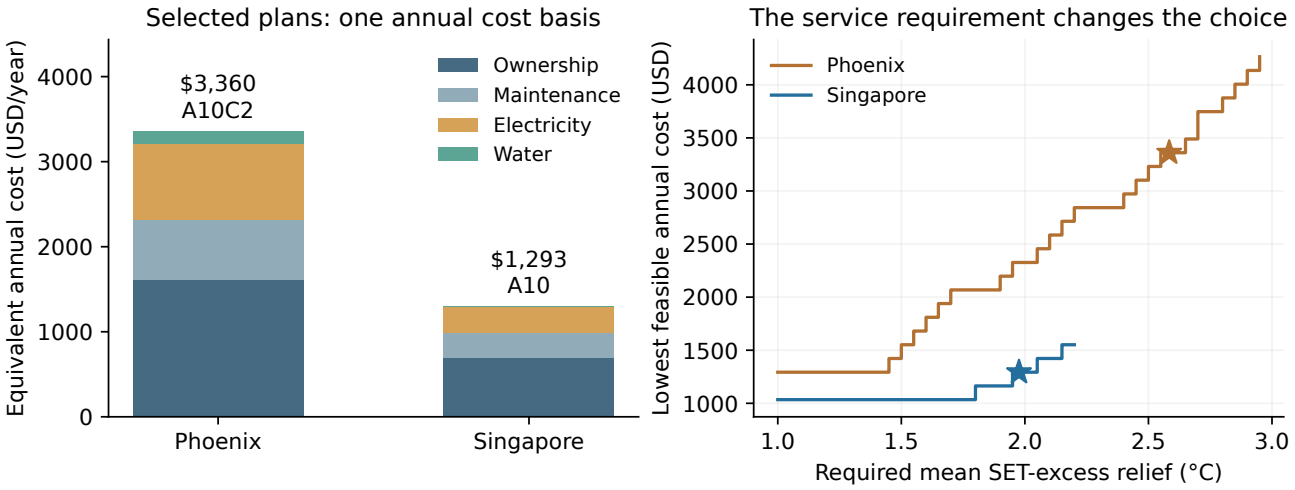


Figure 14: Annual costs of the selected plans and the minimum cost as the required thermal relief increases, holding the other policy limits fixed. Stars mark the stated policy. Stepped curves reflect integer equipment choices.

Memorandum to the Chief Operations Officer

To: Chief Operations Officer, HiMCM
From: Cooling-service modeling team
Subject: A targeted cooling plan for the modeled warm-day dining event

We recommend that HiMCM use **ten type-A fans and two type-C evaporative units** as the planning configuration for the modeled 300-m^2, 120-guest sector at Ullman Terrace. The venue's operations team can implement the layout under the 36°C, 25% RH design condition and the service, access and annual-energy limits in our analysis.

Put both evaporative units and one fan in the busiest zone, five fans in the next, and four in the least crowded zone. This arrangement directs the strongest cooling to the greatest number of guests. The model estimates 96.7% person-time coverage and an average reduction of 2.60°C in SET excess. A four-hour event uses **30 kWh and approximately 319 liters of make-up water**. Its service efficiency is 41.63 person·°C·h/kWh. Remaining SET excess is 4.48°C, so substantial warmth remains despite the improvement.

Under our common price assumptions, initial equipment expenditure is **USD 7,000**; equivalent annual cost is approximately **USD 3,360** at 600 operating hours. This annual figure includes capital recovery, maintenance, electricity and water. Obtain installation and operating quotes before purchase. Fifteen fans alone cost less but miss our specified thermal-relief target. Six C units provide stronger relief but exceed the chosen annual electricity budget.

The installation needs a commissioning check. Confirm that the assumed delivery footprints and air speeds reach seated guests, while equipment, hoses and servicing space leave circulation routes and exits clear. Record conditions near, midway from and at the edge of each representative unit. Recalculate the plan when readings differ from the assumed profiles. If induced air speed is 20% lower, the model selects seven fans and three C units under the same policy. A less effective evaporative delivery profile instead selects six fans and three C units; each trigger is evaluated separately.

Humidity is an important operating trigger. At 35% RH with other Phoenix inputs unchanged, no enumerated layout meets the original thermal target under the unchanged service, access and energy policy. Review timing, shade and the approved service requirement before adding equipment. Follow National Weather Service heat guidance: monitor forecasts, reduce or reschedule strenuous outdoor activity, and provide breaks and access to cooler locations [23].

Public Blog: How Much Comfort Does One Kilowatt-Hour Buy?

An outdoor fan can help people feel cooler while the air thermometer barely changes. An evaporative cooler can lower air temperature while adding moisture. To compare them fairly, we followed the cooling all the way to the guest: what temperature, humidity, air movement and radiant heat reach that person, and how long do they benefit?

Our model uses Standard Effective Temperature, or SET, to combine those influences into one thermal-response measure. We track relief above a stated reference condition, then divide the total person-time benefit by electricity use. One service unit means one person's SET excess is reduced by one degree for one hour. This is a model-based comfort measure, not a degree of outdoor air cooling or a unit of heat extracted.

Consider our Singapore dining scenario: 120 guests, four hours, and ten carefully placed fans. The calculation gives about **971 service units for 10 kWh**, or **97 service units per kWh**. Across all guests, that corresponds to roughly two degrees less SET excess during the event. The fans directly serve about 90% of guest-time.

The Phoenix example needs a different choice to meet its service target. Ten fans plus two evaporative units supply about **1,249 service units for 30 kWh**: about **42 per kWh**. Dry air leaves more room for water to evaporate, but equipment power, delivery losses and where people sit still matter. A higher-powered cooler is worthwhile when its extra relief is needed and reaches enough guests.

One number cannot tell the whole story. Doubling the Singapore dining area and crowd while keeping ten fans leaves the efficiency ratio unchanged, yet halves coverage to 45%. Check who is served, how much warmth remains, and what the equipment costs. During severe heat, follow local advisories and adjust outdoor plans [23, 24].

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[26] Google. *Gemini*, user-supplied analysis excerpt, November 2026. See Report on Use of AI Tools.

Report on Use of AI Tools

OpenAI Codex was used during November 2026 to assist with translation, model formulation, Python implementation, source research, English drafting, LaTeX preparation and figure generation. The user also supplied a Google Gemini analysis as a PDF excerpt; its process-oriented explanation informed the early organization. Gemini was not used as a scientific source for the physical equations or numerical inputs.

Development record. The assistance covered freezer service accounting; psychrometric and SET calculations; the shared local-delivery operator; station heat, moisture and surface balances; candidate deployment comparisons; scope separation between efficiency and effectiveness; and the final CP-SET restructuring. The supplied drawings and earlier award papers were used for context and presentation, with their data provenance distinguished from scenario inputs.

Latest revision. The user requested a named framework, explicit assumptions, a symbol table, preserved core equations and figures, shorter verification prose, an extreme hot-dry test, and a complete paper within 25 counted pages. Codex generated and executed the additional test instead of assuming the requested reversal, retained the computed rankings, and prepared the venue and decision extensions. The model implementation, input files, CSV outputs, figure scripts and verification records are retained with the LaTeX project.

Verification record. Automated checks cover heat and water conservation, service-energy identities, limiting-case recovery, time-step convergence and consistency between tables, figures and outputs. The AI-assisted workflow performed these checks; independent team review is recorded separately by the team.

Tool identification and transcript archive. OpenAI Codex, accessed November 8, 2026; the session identifies Codex as based on GPT-6; a more specific deployment identifier was not supplied. Google Gemini, user-supplied conversation excerpt; the model identifier is not present in that excerpt. The original prompts and complete retained outputs are held in the associated conversation records. This narrative is an activity record, not a replacement for those transcripts.

Policy reference. COMAP, *Use of AI Tools in COMAP Contests*, v102025, `https://www.contest.comap.com/undergraduate/contests/mcm/flyer/Contest_AI_Policy.pdf`; and 2026 HiMCM instructions, `https://www.contest.comap.org/highschool/contests/himcm/instructions.html`, accessed November 8, 2026.